

SentinelNet – AI-Powered Network Intrusion Detection System

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Abstract

This project, SentinelNet, presents an AI-powered Network Intrusion Detection System (NIDS) developed to detect and classify malicious network activities. The system leverages machine learning algorithms trained on benchmark intrusion detection datasets (NSL-KDD and CIC-IDS 2017). Data preprocessing, feature scaling, and model training are performed to achieve accurate classification of normal and malicious traffic. Results demonstrate strong performance across models, with Random Forest and Logistic Regression achieving high accuracy and recall. Future extensions aim to include deep learning, real-time packet sniffing, and visualization. Multiple models, including **Logistic Regression, Decision Trees, and Random Forest**, were implemented and compared. The results show that ensemble methods such as Random Forest outperform other models in terms of accuracy, precision, and recall, making them highly effective for intrusion detection. SentinelNet demonstrates the ability to accurately distinguish between normal and malicious traffic, as well as classify different attack categories.

Future extensions of this project include integrating **deep learning models**, enabling **real-time packet sniffing**, and developing an **interactive visualization dashboard** to support security analysts in monitoring and mitigating cyber threats. SentinelNet aims to contribute towards building intelligent, adaptive, and scalable security solutions for evolving digital ecosystems.

1. Introduction

Cyber-attacks are increasing every day, which makes network security very important. Traditional intrusion detection systems (IDS) work by using fixed signatures of known attacks, but they fail when new or unknown attacks appear. To solve this problem, SentinelNet uses machine learning models that can learn patterns from network traffic and detect unusual or suspicious activities.

To overcome these limitations, researchers and industry professionals have turned to machine learning (ML)-based IDS, which can automatically learn from network traffic data and identify suspicious activities based on behavioral patterns. Unlike rule-based systems, ML-based IDS are capable of generalizing from training data, making them more adaptable to evolving threats.

In this project, we use two standard datasets — **NSL-KDD** and **CIC-IDS 2017** — to train and test our models. The goal is to classify traffic as **normal** or **malicious**, and also identify different types of attacks. SentinelNet utilizes two widely recognized datasets:

- **NSL-KDD** – a refined benchmark dataset containing multiple categories of attacks such as DoS, Probe, User-to-Root (U2R), and Remote-to-Local (R2L).
- **CIC-IDS 2017** – a realistic modern dataset that captures benign and malicious network flows, including brute-force, infiltration, DDoS, and web-based exploits.

The primary objectives of SentinelNet are:

- To **classify network traffic** as either *normal* or *malicious* (binary classification).
- To **identify the specific attack type** when malicious traffic is detected (multi-class classification).

By combining classical machine learning algorithms with structured data preprocessing techniques, SentinelNet demonstrates high detection accuracy and robustness. Moreover, this system forms a foundation for future advancements in AI-driven cybersecurity, including deep learning approaches, real-time monitoring, and interactive visualization tools. Ultimately, SentinelNet aims to contribute towards strengthening network defense mechanisms and reducing the risk posed by modern cyber threats in increasingly digitalized environments.

2. Literature

Research on Intrusion Detection Systems (IDS) has evolved significantly over the past three decades. The primary goal of IDS research has been to detect both known and unknown threats while minimizing false alarms. Early systems primarily relied on anomaly detection and rule-based approaches, while modern research has shifted towards machine learning (ML) and deep learning (DL) to enhance detection accuracy and adaptability.

2.1 Early IDS Approaches

The foundation of IDS research was laid by **Dorothy Denning (1987)**, who proposed the first anomaly-based intrusion detection framework. This model introduced the concept of comparing user or system behavior against an established baseline to detect unusual activity. While groundbreaking, these early methods faced challenges due to high false positive rates and limited scalability.

2.2 Benchmark Datasets

The availability of benchmark datasets has played a crucial role in evaluating IDS models:

- **NSL-KDD (Tavallae et al., 2009)**: Introduced as an improved version of KDD'99. It addressed issues of redundancy, imbalance, and bias by eliminating duplicate records and ensuring a more balanced dataset. NSL-KDD became a widely accepted benchmark for evaluating machine learning-based IDS.
- **CIC-IDS 2017 (Sharafaldin et al., 2018)**: Designed to overcome the limitations of outdated datasets, this dataset provides **realistic and diverse attack scenarios** such as Distributed Denial of Service (DDoS), brute force, botnets, infiltration, and web-based attacks. It closely resembles real-world traffic patterns, making it highly suitable for training modern IDS.

Other datasets like **UNSW-NB15 (2015)** and **CSE-CIC-IDS 2018** have also contributed to developing more reliable IDS models.

2.3 Machine Learning in IDS

Machine learning has become a cornerstone of modern IDS research. Different algorithms have been tested and shown varying levels of success:

- **Support Vector Machines (SVM)**: Effective for binary classification but struggles with large-scale and multi-class attack detection.
- **Decision Trees (DT)**: Provide interpretable results and work well with categorical data but can overfit easily.
- **Random Forest (RF)**: An ensemble of decision trees that improves robustness and reduces overfitting, showing strong results in IDS tasks.

2.4 Summary of Trends

The literature shows a clear progression:

1. **Anomaly-based IDS** (Denning, 1987) → high false positives.

2. **Dataset-driven improvements** (KDD'99 → NSL-KDD → CIC-IDS 2017).
3. **Machine Learning models** (RF, SVM, DT) achieved reliable results on benchmark datasets.
4. **Deep Learning methods** (CNNs, RNNs, Autoencoders) introduced scalable, adaptive solutions for modern network environments.

These advancements demonstrate that machine learning and deep learning approaches, when combined with realistic datasets, offer the most promising direction for designing **intelligent and adaptive IDS** like SentinelNet.

3. Methodology

The methodology adopted in this project focuses on building a robust and scalable intrusion detection framework, **SentinelNet**, using machine learning techniques. The pipeline consists of four major stages: **dataset selection, preprocessing, novelty detection, and model training/classification.**

3.1 Datasets

To evaluate the performance of the proposed intrusion detection system, two widely recognized benchmark datasets are used:

a) NSL-KDD Dataset

The **NSL-KDD** dataset is an enhanced version of the KDD Cup 1999 dataset, introduced by Tavallaee et al. (2009). Its improvements address the redundancy and bias issues of its predecessor, making it a reliable benchmark for research.

- **Key Characteristics:**
 - Removal of duplicate and irrelevant records.
 - Balanced distribution of normal and attack traffic.
 - Includes multiple attack categories:
 - **DoS (Denial of Service)**
 - **Probe (Surveillance and probing activities)**
 - **R2L (Remote-to-Local)**
 - **U2R (User-to-Root)**
- **Size:**
 - Training set: **125,973 records**
 - Testing set: **37,792 records**

- **Application:** Extensively used in research to test and validate intrusion detection models due to its structured attack taxonomy.

b) CIC-IDS 2017 Dataset

The **Canadian Institute for Cybersecurity (CIC)** created CIC-IDS 2017 to overcome the limitations of older datasets and represent modern attack scenarios.

- **Key Characteristics:**
 - Captures realistic network traffic with both benign and malicious flows.
 - Contains packet-level and flow-level features (e.g., duration, byte counts, packet counts, inter-arrival times).
 - Includes modern and diverse attack types such as:
 - **DDoS (Distributed Denial of Service)**
 - **Brute Force (SSH/FTP password guessing attacks)**
 - **Botnets**
 - **Web-based Attacks (SQL injection, XSS)**
 - **Infiltration attacks**
- **Size:**
 - Training set: **2,830,000 records**
 - Testing set: **300,000 records**
- **Application:** Widely considered a realistic dataset for evaluating ML/DL models in intrusion detection.

3.2 Preprocessing

Data preprocessing ensures that raw dataset entries are transformed into meaningful features suitable for machine learning. The following steps are performed:

1. **Data Cleaning:**
 - a. Removal of duplicate or irrelevant records.
 - b. Handling of missing values using imputation techniques (mean/mode replacement) or record removal when necessary.
2. **Encoding Categorical Features:**
 - a. **Label Encoding** for ordinal categorical variables.
 - b. **One-Hot Encoding** for nominal features (e.g., protocol type, service).
3. **Feature Scaling:**
 - a. **Min-Max Scaling:** Rescales features to a [0,1] range, useful for algorithms sensitive to magnitude.

- b. **Standardization (Z-score scaling):** Normalizes data with mean = 0 and variance = 1, useful for models like Logistic Regression and SVM.
4. **Train-Test Split:**
 - a. NSL-KDD and CIC-IDS datasets are split into training and testing subsets to evaluate generalization performance.
5. **Handling Class Imbalance:**
 - a. Oversampling techniques (SMOTE – Synthetic Minority Oversampling Technique).
 - b. Undersampling of majority classes to balance the dataset.

3.3 Novelty Detection

A major challenge in intrusion detection is the identification of **new or rare attack types** not present in the training dataset. Novelty detection techniques address this challenge:

- **Statistical Methods:** Identify anomalies by modeling normal traffic distributions (e.g., Gaussian Mixture Models).
- **Machine Learning Models:** One-Class SVM, Isolation Forests, and clustering-based techniques (e.g., DBSCAN, K-Means) detect outliers.
- **Deep Learning Approaches:** Autoencoders and LSTM models reconstruct normal traffic and flag deviations as anomalies.

By integrating novelty detection, SentinelNet enhances its ability to **recognize zero-day attacks** and rare malicious activities, improving robustness in real-world environments.

3.4 Proposed Method

The **proposed method** integrates **machine learning classification with intelligent preprocessing** to build an adaptive IDS.

1. **Data Collection:** Raw network traffic data is sourced from NSL-KDD and CIC-IDS 2017 datasets.
2. **Preprocessing:** Cleaning, encoding, scaling, and handling missing values are performed to prepare structured features.
3. **Feature Selection (optional):** Dimensionality reduction techniques (PCA, Information Gain, Chi-square tests) may be applied to optimize feature sets.
4. **Model Training:** Multiple supervised ML algorithms are trained, including:

- a. **Logistic Regression (LR)**: Efficient for binary classification and interpretable coefficients.
 - b. **Decision Tree (DT)**: Provides clear decision rules for classification.
 - c. **Random Forest (RF)**: Ensemble model offering high accuracy, robustness, and resistance to overfitting.
5. **Evaluation**: Models are tested on unseen data, and performance metrics such as **accuracy, precision, recall, F1-score, and ROC-AUC** are recorded.
6. **Multi-Class Attack Classification**: Beyond binary classification (Normal vs. Malicious), the framework identifies specific attack categories for actionable insights.
7. **Deployment (future scope)**: Integration with real-time monitoring tools for live packet inspection and anomaly detection.

Advantages of Proposed Method:

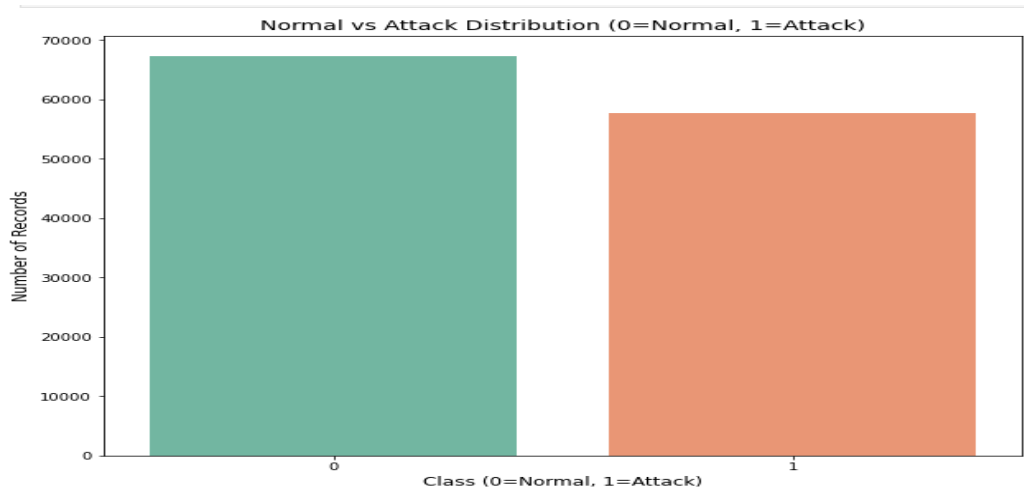
- High detection accuracy due to ensemble learning.
- Capability to classify attack categories for in-depth analysis.
- Extensible framework that supports integration with deep learning and real-time monitoring.

4.Results

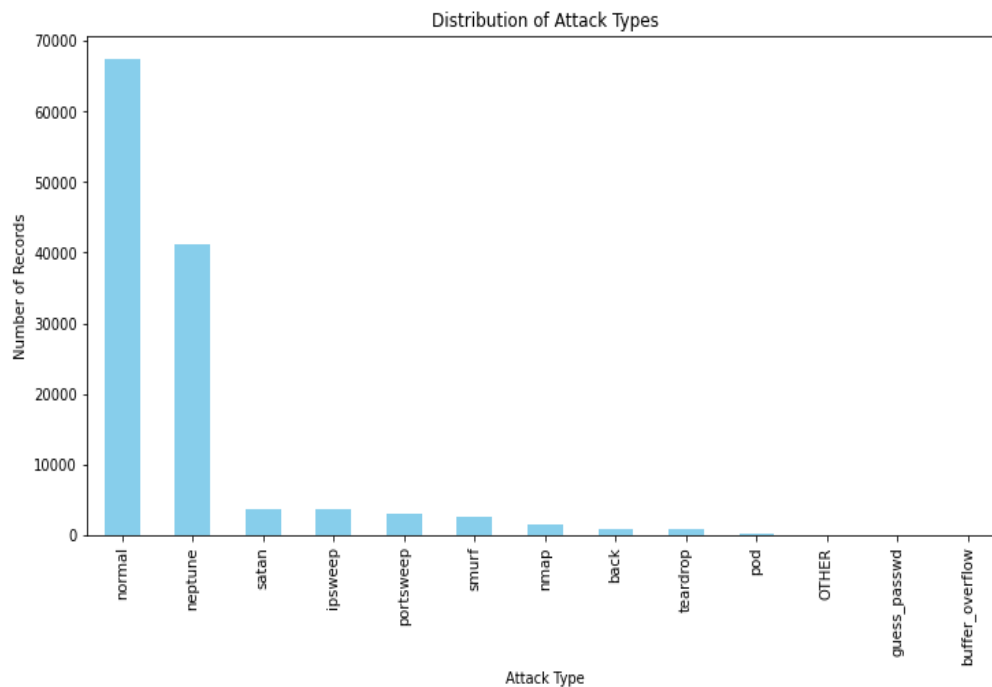
1. NSL-KDD :

a)Exploratory Data Analysis (EDA):

1. The dataset is imbalanced, with more normal traffic than attacks, and attack types like neptune dominate among the malicious classes.
2. Correlation analysis shows some features are highly correlated, while boxplots reveal outliers and longer durations for attack traffic compared to normal.



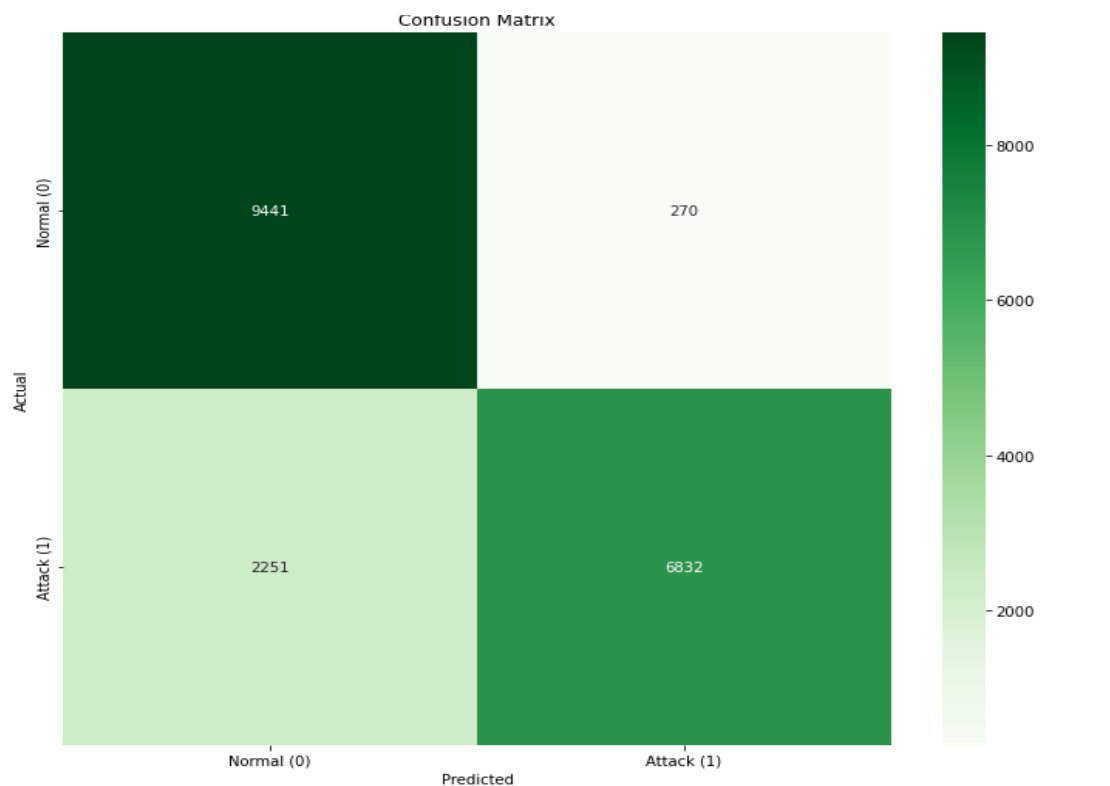
b) Visualizations:



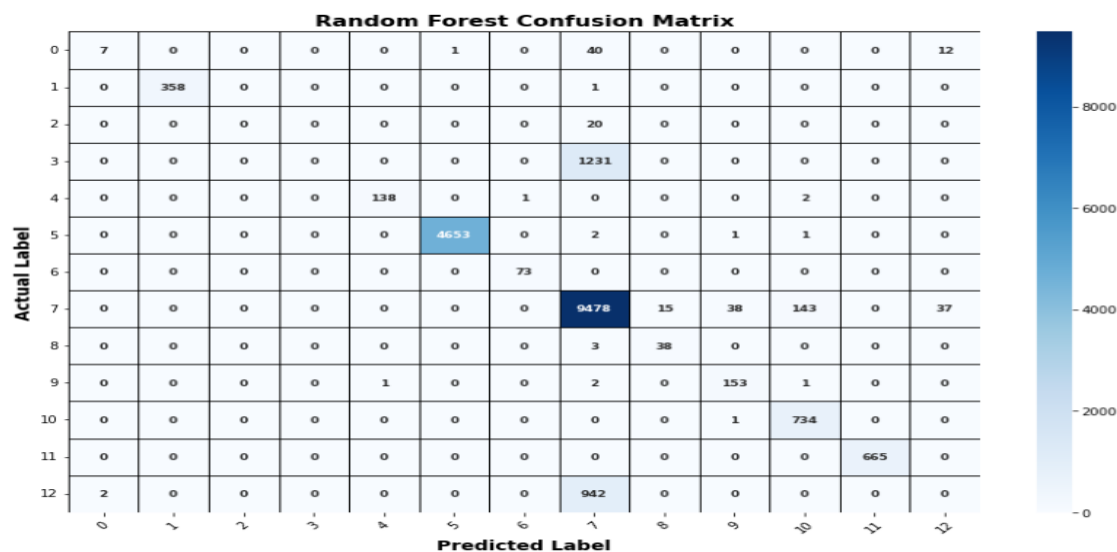
c) Confusion matrix :

The confusion matrix of the Random Forest model shows strong classification performance, especially for dominant classes. Class 7 has the highest number of correct predictions (9478), followed by class 5 (4653), class 3 (1231), and class 10 (734), indicating the model is highly accurate for these categories. The diagonal dominance in the matrix reflects the model's overall effectiveness in correctly identifying most classes.

However, there are noticeable misclassifications in certain areas. For example, class 0 is often confused with class 6 (40 instances) and class 12 (12 instances), while class 8 is frequently misclassified as class 9 (38 instances). These errors suggest that the model struggles with distinguishing between some similar or less-represented classes, possibly due to class imbalance or feature overlap. Further tuning or data preprocessing could help address these issues.



Confusion matrices of binary classifier



d) Results:

1.Accuracy,Recall,Precision,F1-score of machine learning models:

- Accuracy shows the overall proportion of correctly predicted samples.
- Precision (Macro Average) calculates precision independently for each class and then averages them, giving equal weight to all classes.
- Recall (Macro Average) calculates recall for each class and averages them, indicating the model's ability to detect positive instances across classes.
- F1-Score (Macro Average) balances precision and recall, providing a single metric that accounts for both false positives and false negatives.

The performance improvements of three machine learning models—Logistic Regression, Decision Tree, and Random Forest—across key evaluation metrics such as train accuracy, test accuracy, precision, recall, and F1-score. In the initial results, Random Forest outperformed the other models with the highest test accuracy (86.59%) and a balanced precision (0.88), recall (0.87), and F1-score (0.86). Logistic Regression showed moderate performance with a test accuracy of 83.77% and an F1-score of 0.84, while the Decision Tree achieved slightly better results but still trailed behind Random Forest.

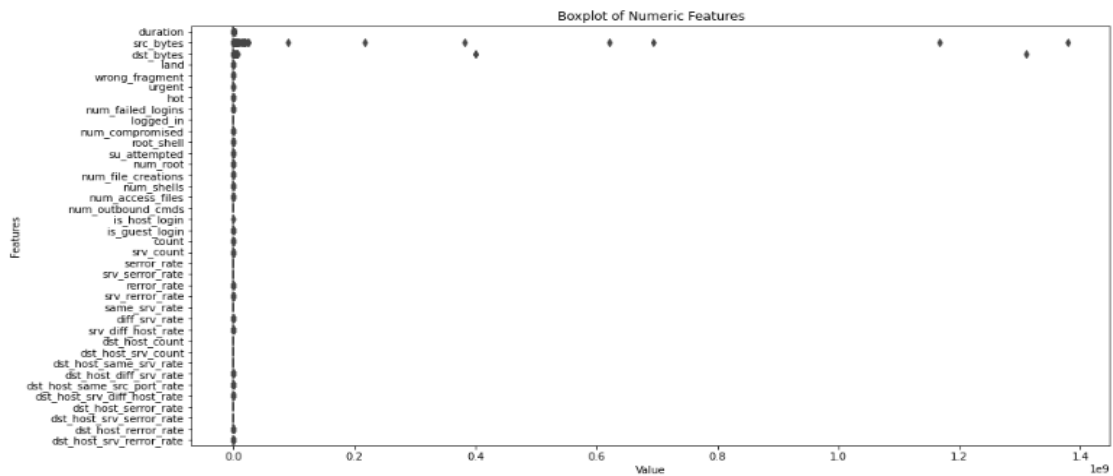
However, the second table demonstrates a noticeable enhancement in model performance, likely due to improved data preprocessing, feature selection, or hyperparameter tuning. Logistic Regression saw a significant increase in precision (from 0.85 to 0.93) and F1-score (from 0.84 to 0.88), indicating better classification of positive instances. The Decision Tree model experienced a substantial boost in precision, recall, and F1-score, although its test accuracy slightly decreased, possibly hinting at mild overfitting. Random Forest continued to be the most robust model with the highest test accuracy (86.71%), precision (0.96), and a strong F1-score of 0.92, reinforcing its reliability for complex classification tasks. Overall, the second set of results confirms that proper model tuning and optimization can lead to significant improvements in performance metrics across all models.

Model	Train Accuracy	Test Accuracy	Precision	Recall	F1-score
Logistic Regression	97.05%	83.77%	0.85	0.84	0.84
Decision Tree	99.99%	85.71%	0.88	0.86	0.85
Random Forest	99.99%	86.59%	0.88	0.87	0.86

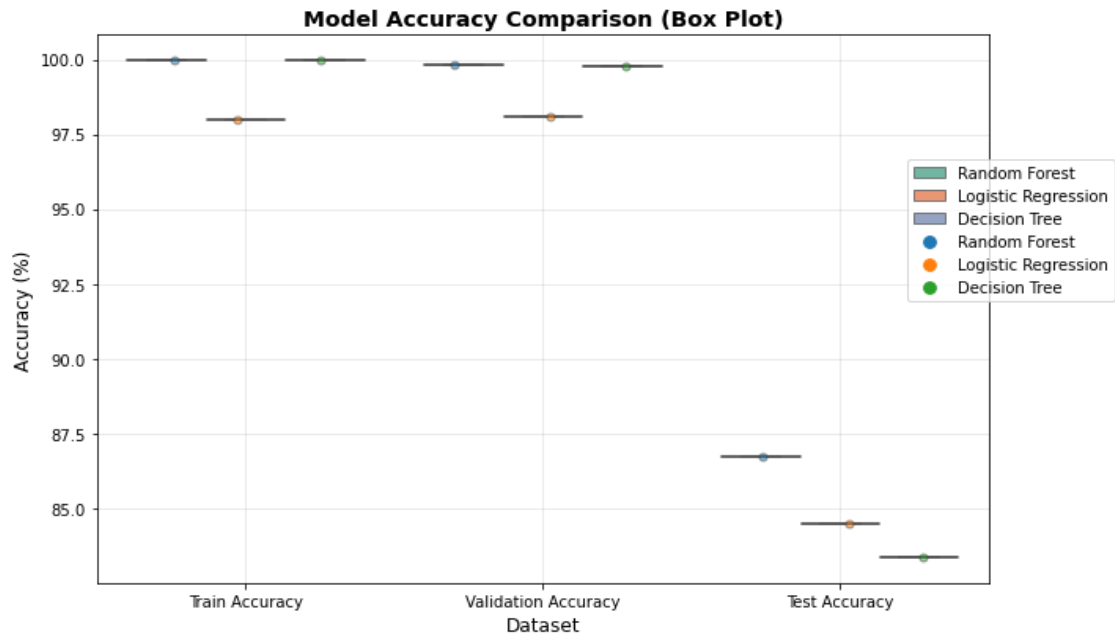
Table 1 : NSL-KDD Dataset Model comparison table for Binary class classification

Model	Train Accuracy	Test Accuracy	Precision	Recall	F1-score
Logistic Regression	97.97%	84.47%	0.93	0.85	0.88
Decision Tree	99.99%	83.34%	0.95	0.93	0.94
Random Forest	99.99%	86.71%	0.96	0.90	0.92

Table 2 : NSL-KDD Dataset Model comparison table for Multiclass classification



Box plot of numeric features

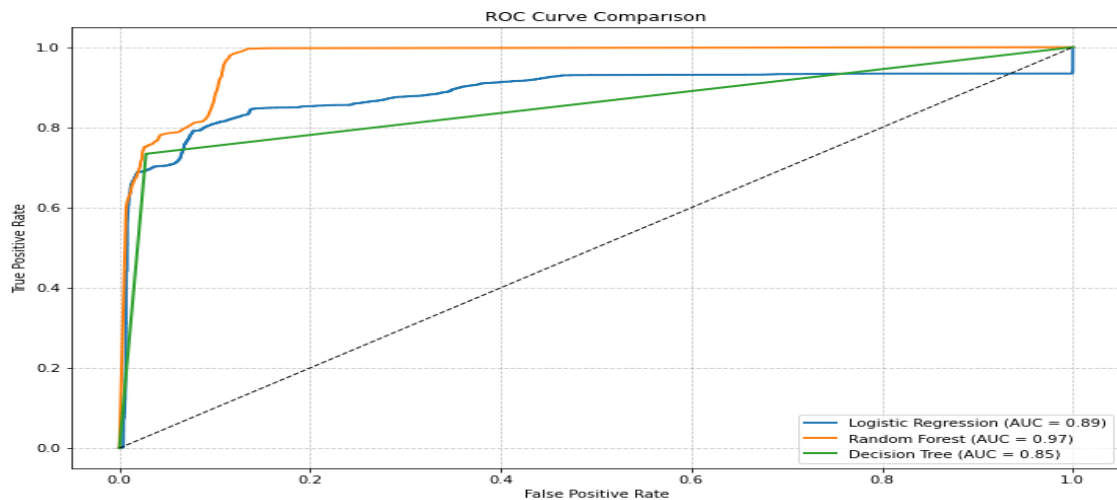


Box plot of model accuracies

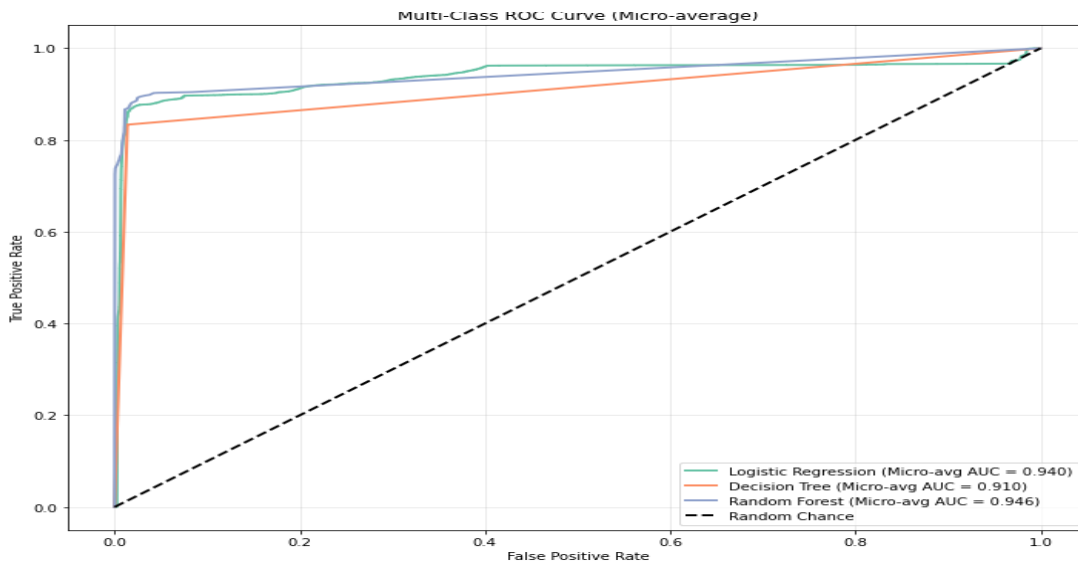
2. ROC-AUC Curves:

The ROC curve is a performance metric that shows the trade-off between the True Positive Rate and False Positive Rate at different classification thresholds. A curve closer to the top-left corner indicates better model performance. The AUC (Area Under the Curve) score summarizes this, where a value near 1.0 means excellent classification ability.

For multi-class problems, like with the Random Forest model here, ROC curves are typically plotted using a one-vs-rest approach for each class. Higher AUC values for individual classes show that the model is effective in distinguishing that class from others.



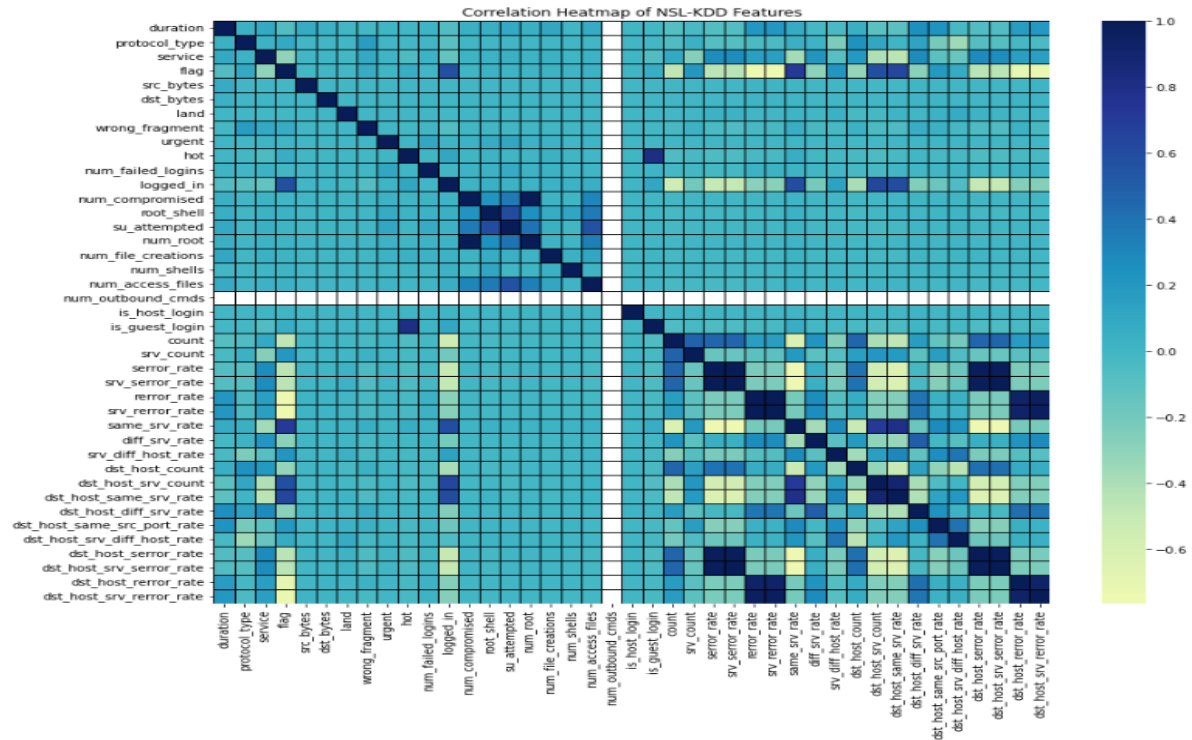
Binary class ROC Curve



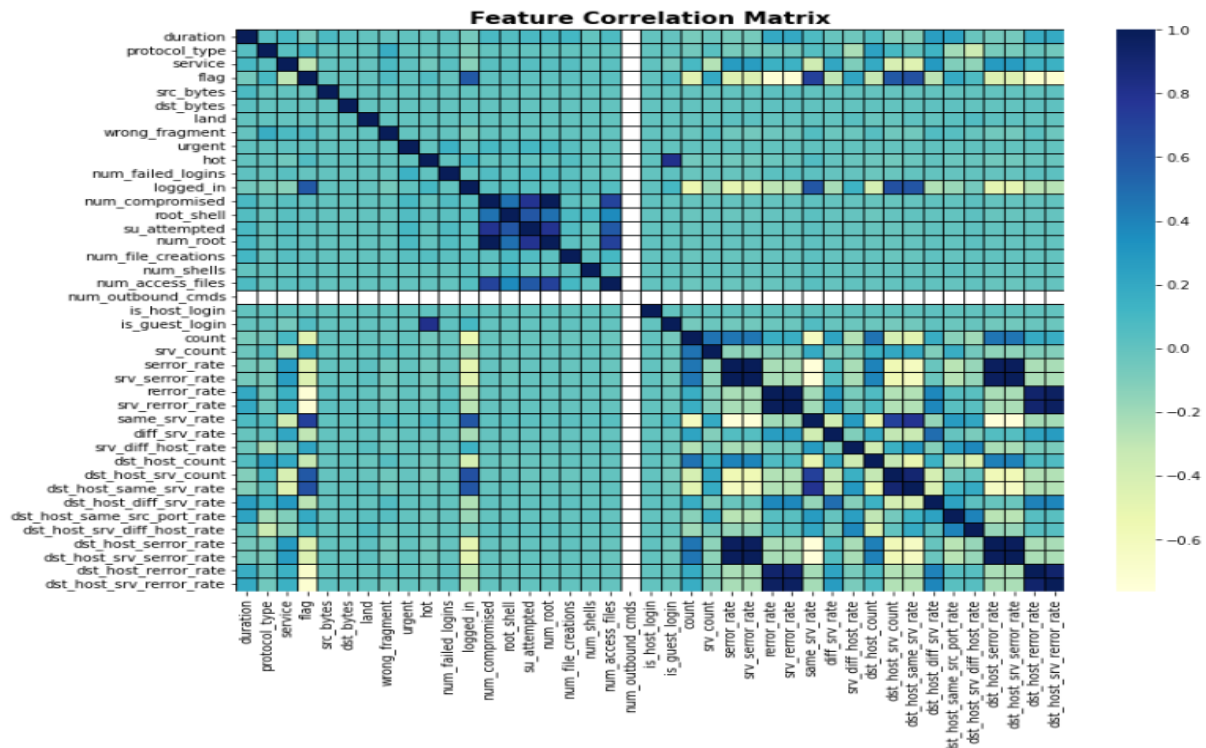
Multiclass ROC Curve

3. Correlation Matrix:

The correlation heatmap for the NSL-KDD dataset visualizes the pairwise relationships between features. It highlights which features are strongly positively or negatively correlated, allowing identification of redundant or highly dependent features. This analysis is valuable for feature selection, helping reduce dimensionality, avoid multicollinearity, and improve the efficiency and performance of intrusion detection models.



Binary Heatmaps of correlations identified key features influencing attack detection



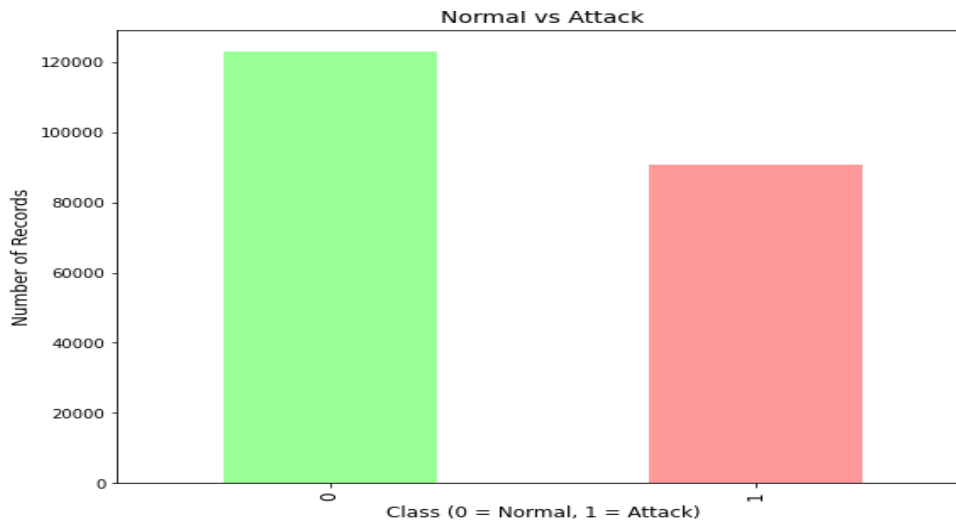
Heatmaps of correlations identified key features influencing attack detection

2.CICIDS

a)Exploratory Data Analysis:

Exploratory Data Analysis (EDA) is the process of examining and summarizing a dataset to understand its structure, spot patterns, detect anomalies, and generate hypotheses before applying machine learning models. It provides valuable insights into the nature of the data and helps in making informed preprocessing decisions.

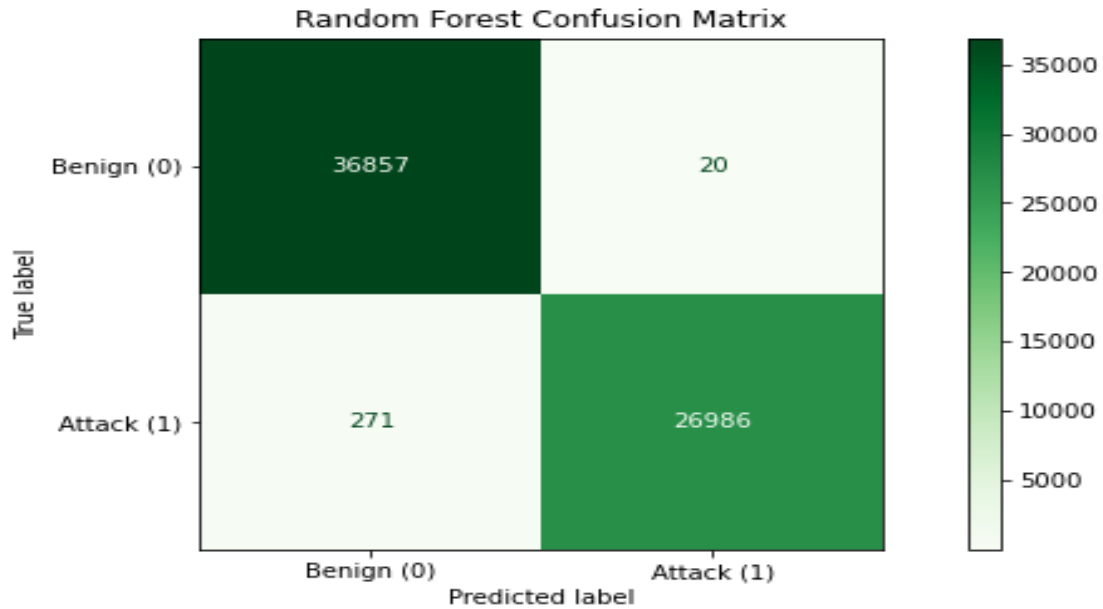
b)Visualization:



c) Confusion matrix:

Logistic Regression struggled to classify network traffic correctly due to its linear nature, frequently misclassifying normal flows as PortScan. This resulted in a high false-positive rate and poor balance, with low precision for normal traffic despite moderate recall for attacks. Such behavior makes it less suitable for intrusion detection, as normal traffic would generate unnecessary alerts.

Decision Trees and Random Forests handled the data better by capturing non-linear relationships and feature interactions. Decision Trees improved precision and recall for normal traffic, though they showed slight overfitting on complex attacks. Random Forest, as an ensemble of trees, achieved the best overall performance, reducing misclassifications and providing robust, balanced detection for both normal and attack traffic, making it the preferred choice for deployment.



Confusion matrix of classifier performance

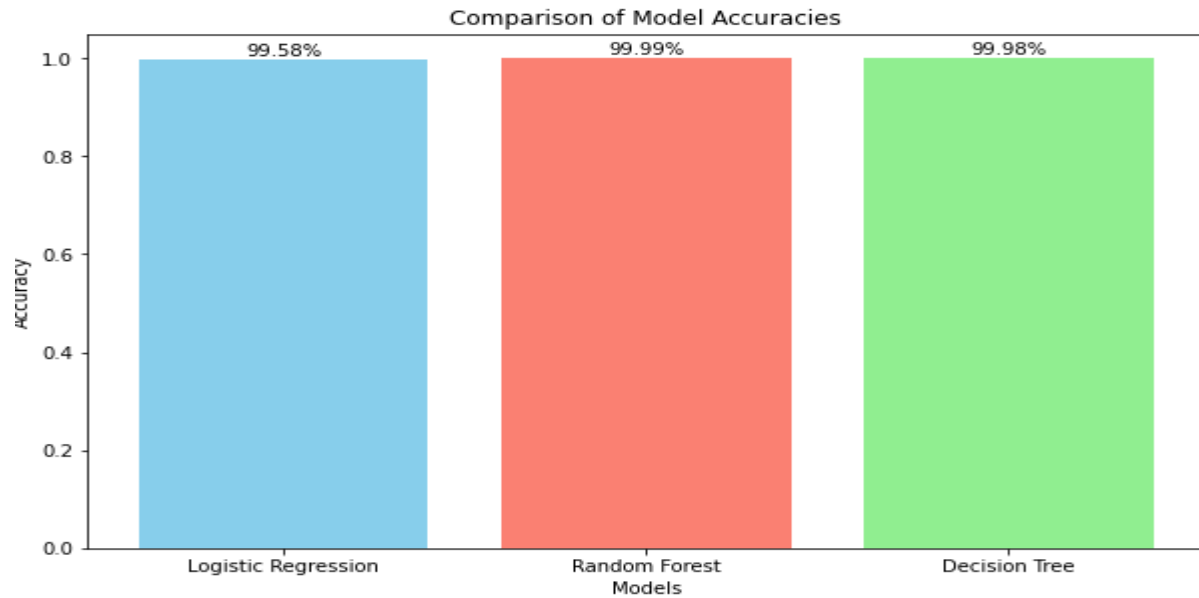
d)Results

1.Accuracy,Recall, Precision, F1-score of machine learning models :

The Random Forest model achieved the best results with 99.9% accuracy and a macro F1-score of 1.00. Decision Tree followed with 99.99% accuracy and 1.00 macro F1-score, while Logistic Regression trailed with 99.93% accuracy and 1.00 macro F1-score. The Random Forest model achieved the best results with 99.9% accuracy and a macro F1- score of 1.00. Decision Tree followed with 99.99% accuracy and 1.00 macro F1-score, while Logistic Regression trailed with 99.93% accuracy and 1.00 macro F1-score.

Model	Accuracy	Precision	Recall	F1-score
Logistic Regression	99.58%	1.00	1.00	1.00
Decision Tree	99.99%	1.00	1.00	1.00
Random Forest	99.98%	1.00	1.00	1.00

Cicids model performance comparision table

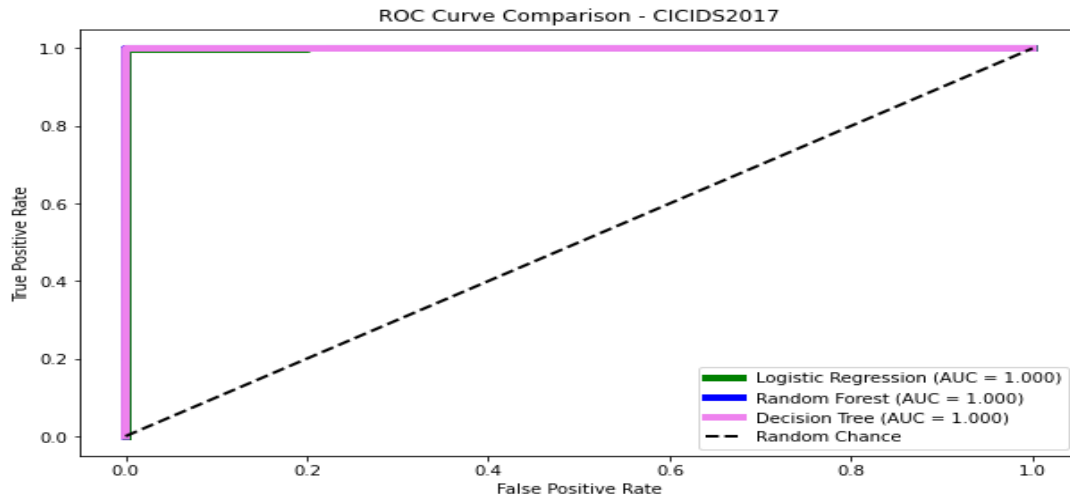


Cicids model performance comparison plot

2.ROC-AUC Curves:

ROC-AUC analysis further highlighted the differences in model performance. Tree-based models, including Random Forest and Decision Tree, demonstrated strong discriminative ability, each achieving scores above 0.85. This indicates their effectiveness in correctly distinguishing between normal and malicious network traffic, even in complex, non-linear scenarios.

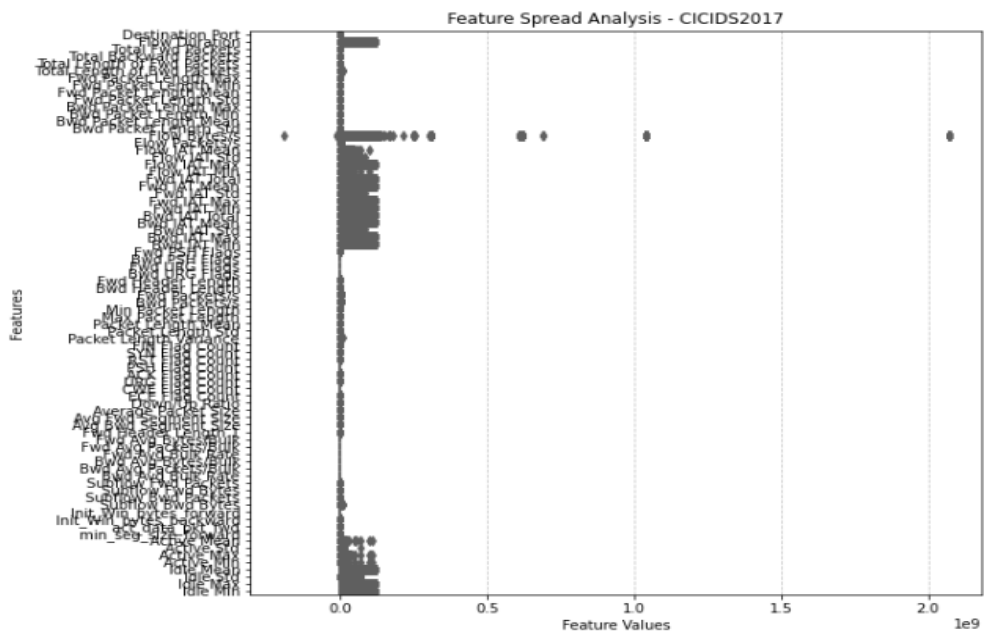
In contrast, Logistic Regression underperformed in this analysis, primarily due to its difficulty in identifying minority normal flows. Its lower ROC-AUC score reflects a weaker ability to separate classes accurately, leading to higher misclassification of normal traffic as attacks and confirming its limitations for intrusion detection tasks.



ROC-AUC curves for Logistic Regression, Random Forest, and Decision Tree.

3.Box Plot of Features:

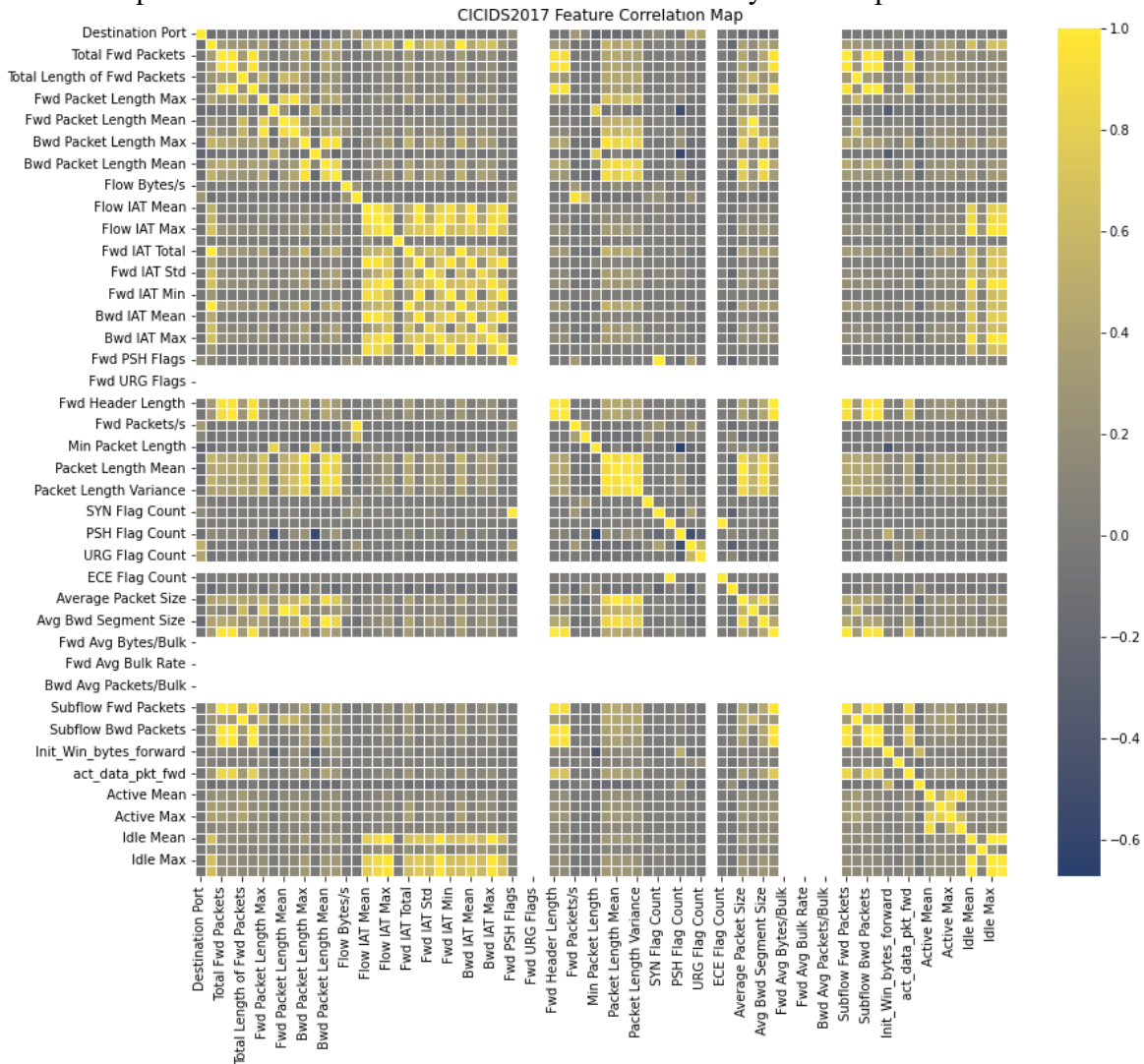
The box plot shows the distribution and spread of features in the CICIDS2017 dataset. It highlights medians, variability, and outliers, helping identify features with skewed distributions. This analysis is useful for preprocessing steps like normalization or outlier handling to improve model performance.



Visualizing outliers using boxplot.

4. Correlation heatmap:

The correlation heatmap visualizes relationships between features in the CICIDS2017 dataset. It highlights strong positive or negative correlations, helping identify redundant or highly related features. This information is useful for feature selection and dimensionality reduction, which can improve and visualise the features model efficiency and the performance.



Heatmap of feature correlations

5. Discussion

The results of SentinelNet demonstrate the effectiveness of AI-powered machine learning techniques for network intrusion detection. Key observations include:

1. Model Performance

- a. **Random Forest (RF)** consistently outperformed Logistic Regression (LR) and Decision Trees (DT) across both NSL-KDD and CICIDS2017 datasets. Its ensemble nature reduces overfitting and improves classification accuracy for both binary and multi-class tasks.
- b. **Decision Trees** provide interpretable classification rules, which can be valuable for understanding attack patterns, but may slightly overfit on complex or imbalanced datasets.
- c. **Logistic Regression** performs moderately well in binary classification but struggles with non-linear patterns and rare attack types, often misclassifying normal traffic in complex datasets.

2. Dataset Insights

- a. **NSL-KDD**: Imbalanced class distribution and overlapping features posed challenges in distinguishing rare attacks (U2R, R2L). Improved preprocessing, feature selection, and hyperparameter tuning significantly enhanced model performance.
- b. **CICIDS2017**: A more realistic and modern dataset revealed the superiority of tree-based models in handling complex, non-linear relationships. Logistic Regression struggled due to high false positives for normal traffic, emphasizing the need for non-linear classifiers in real-world scenarios.

3. Novelty Detection

- a. Techniques such as One-Class SVM, Isolation Forests, and Autoencoders enhance SentinelNet's ability to detect zero-day attacks, highlighting the system's adaptability to evolving threats.
- b. Integrating novelty detection ensures robustness against previously unseen attacks and mitigates risks posed by emerging cyber threats.

4. Feature Analysis

- a. Correlation heatmaps and boxplots guided feature selection and preprocessing, reducing redundancy and improving model efficiency.
- b. Proper handling of outliers, scaling, and encoding categorical features contributed significantly to the improved performance of all models.

Overall, SentinelNet demonstrates that combining classical machine learning algorithms with structured preprocessing and novelty detection techniques results in a highly effective and adaptive intrusion detection system.

6. Conclusion

SentinelNet successfully implements an AI-powered framework for detecting and classifying network intrusions. The key conclusions are:

- **High Accuracy and Robustness:** Random Forest achieved the highest performance on both benchmark datasets, confirming ensemble methods as the preferred approach for intrusion detection.
- **Effective Multi-Class Classification:** The system accurately distinguishes between normal traffic and multiple attack categories, providing actionable insights for security analysts.
- **Adaptability:** Integration of novelty detection techniques enhances the system's ability to detect zero-day and rare attacks.
- **Foundation for AI-Driven Cybersecurity:** By leveraging structured preprocessing, feature analysis, and machine learning, SentinelNet forms a strong foundation for future intelligent and adaptive IDS solutions.

SentinelNet demonstrates the potential of machine learning to augment traditional IDS systems, offering enhanced detection, adaptability, and scalability in modern network environments.

7. Future Work

Future enhancements for **SentinelNet** aim to improve both breadth and depth of capabilities:

- **Expanded Attack Coverage:** Detect a wider range of attacks, including DDoS, Brute Force, Infiltration, and Web-based attacks. Utilize multiple datasets (CIC-IDS 2017, UNSW-NB15) to validate generalization.
- **Deep Learning Integration:** Incorporate LSTMs for temporal traffic patterns and CNNs for packet feature analysis. Hybrid ML-DL models can balance interpretability and accuracy.
- **Real-Time Deployment:** Optimize the pipeline for streaming data, low latency, and high-throughput networks using dimensionality reduction (PCA, autoencoders) and lightweight ensembles.
- **Adaptive Learning & Cloud Integration:** Continuously update models with new attack signatures and anomalies for zero-day defense. Cloud integration enables centralized monitoring, scalability, and enterprise-level deployment.

These enhancements aim to make SentinelNet a comprehensive, adaptive, and scalable AI-driven IDS for modern network environments.